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# Within and among years atmospheric moisture availability influences wildfire burn severity in the boreal forest

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## Abstract

**Background** Atmospheric moisture can either mitigate or exacerbate burn severity depending on fuel and environmental conditions. However, changes in fuel moisture typically lag behind shifts in atmospheric moisture associated with weather or climate changes. When atmospheric moisture decreases, lagged decreases in fuel moisture can affect fire hazard in two ways: they can increase fuel flammability and wildfire severity in the short term, or alternatively, they can decrease wildfire severity over longer periods by slowing the accumulation of combustible materials. Yet, the temporal scales at which lagged moisture changes most strongly influence burn severity are not well understood. Hence, we investigated how time-lagged moisture availability at both inter- and intra-annual timescales, forest composition, and drought conditions during the burn period contribute to wildfire burn severity across 1025 fires (1986–2024) in Ontario (Canada). Specifically, we quantified the relative effects of daily vapour pressure deficit (VPD), annual climate moisture index (CMI) anomalies, fuel composition, and fire-season weather using distributed lag generalized additive models and hierarchical partitioning to compare the contributions of moisture at intra-annual lag intervals up to 30 days before fire (VPD) and interannual lag intervals up to 5 years before fire (CMI).

**Results** We found that intra-annual atmospheric dryness and interannual moisture availability jointly regulate burn severity, with effects varying by lag. Coniferous forest cover was the strongest predictor of both median and extreme severity. Interannual moisture availability (CMI) explained more variation than intra-annual drying (VPD). Wetter-than-normal conditions 4–5 years before fire increased severity, while long-lag dryness reduced it, likely through fuel limitations. At 1–2 year lags, dry conditions increased severity and wet conditions lowered it. When considering extreme burn severity, cumulative multi-day VPD was more influential than individual lag effects.

**Conclusions** These results show that burn severity is driven by short-term flammability and multi-year fuel legacies, underscoring the need to integrate lagged moisture dynamics into fire-severity modelling under climate change.

**Keywords** Climate change, Boreal forest, Lagged effects, Ontario, Boreal shield, Climate moisture index, Relativized burn ratio, Burn severity

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## Resumen

**Antecedentes** La humedad atmosférica puede tanto mitigar como exacerbar la severidad de los incendios, dependiendo de los combustibles y de las condiciones ambientales. Sin embargo, los cambios en la humedad de los combustibles ocurren típicamente después de que se producen en la atmósfera, asociados al tiempo meteorológico o cambios en el clima. Cuando la humedad atmosférica decrece, las disminuciones posteriores en la humedad del combustible pueden afectar el riesgo de incendio de dos maneras: pueden incrementar la inflamabilidad del combustible y la severidad del incendio a corto plazo, o, alternativamente, puede hacer decrecer la severidad del fuego en períodos más largos mediante una más lenta acumulación del material combustible. Aún así, la escala temporal en la cual la demora en los cambios en la humedad influyen más fuertemente la severidad del fuego, no están bien comprendidas. Por ese motivo, estudiamos cómo la demora en la disponibilidad de la humedad en una escala temporal a niveles tanto intra-anual como inter anuales, la composición del bosque, y las condiciones de sequía durante el período de ocurrencia del incendio, contribuyeron a la severidad del fuego en 1025 incendios ocurridos entre 1986 y 2024 en Ontario, Canadá. Específicamente, cuantificamos los efectos relativos del déficit de presión de difusión del vapor diario (VPD), las anomalías registradas en el índice anual de humedad del clima (CMI), la composición del combustible, y el tiempo meteorológico durante la estación de incendios, usando modelos aditivos generalizados retrasados en el tiempo y partición jerárquica, para comparar las contribuciones de humedad a intervalos con diferencias intra-anales de hasta 30 días antes de los incendios (DPV) y los tiempos de intervalo interanuales de hasta 5 años previos al fuego (CMI).

**Resultados** Encontramos que tanto la sequedad intra-anual de la atmósfera como la disponibilidad interanual de esa humedad regulan juntos la severidad de los incendios, con los efectos que varían de acuerdo al retraso entre esos lapsos de ocurrencia. La cobertura de los bosques de coníferas fue el predictor más fuerte tanto de la mediana como de extremos en la severidad de los incendios. La disponibilidad anual de humedad (CMI) explica una mayor variación que el secado intra-anual (VPD). Condiciones de humedad más altas que las normales 4-5 años antes del fuego incrementan su severidad, mientras que el retraso en largos períodos de sequía reducen la severidad, probablemente por limitaciones en la disponibilidad de material combustible. En intervalos de entre 1 -2 años, las condiciones de sequía incrementan la severidad y condiciones húmedas la disminuyen. Cuando consideramos severidad extrema del fuego, la acumulación de días en el VPD tuvo más influencia que los efectos individuales de su retraso en el tiempo.

**Conclusiones** Estos resultados muestran cómo la severidad de los incendios es conducida fundamentalmente por la inflamabilidad en el corto plazo y lo que representa el legado de muchos años de acumulación de combustibles, subrayando la necesidad de integrar el tiempo de demora que ocurre en la dinámica de la humedad, en los modelos de severidad del fuego bajo el paradigma del cambio climático.

## Introduction

Wildfire activity in boreal forests is increasing in response to climate warming (Hanes et al. 2019), with significant negative effects on forest resilience and ecosystem services provisioning (Gauthier et al. 2014). Burn severity, the ecological impacts of fire on vegetation and soil (Keeley 2009), shapes forest ecosystem structure and function by determining the state of the ecosystem immediately following fire. The processes of forest colonization, growth, and succession that drive post-fire recovery are regulated by the effects of burn severity on seed availability, physical floor properties, and soil nutrient status (Johnstone and Kasischke 2005; Miesel et al. 2015). Although wildfire activity and risk are increasing, there is no consistent national trend in burn severity across Canada's boreal forest over the three decades from 1985 to 2015 (Guindon et al. 2021). Yet, models predict spatial variability in burn severity due to variation in moisture conditions, vegetation

and fuel load (Guindon et al. 2021; Wang et al. 2022; Whitman et al. 2018). An important outstanding question is the temporal scale at which changes in moisture conditions, from short-term variability in atmospheric dryness to long-term moisture availability, shape the occurrence of high-severity fire.

Ongoing climate warming and increases in drought conditions are projected to produce more extreme wildfire seasons in Canada (Wang 2024). Evidence of this trend is already apparent, as 2023 saw a record-breaking 15 Mha burned, surpassing the previous record of 7.6 Mha set in 1989 (Jain et al. 2024; Skakun et al. 2022). In addition to increases in area burned, future wildfires are expected to burn with greater intensity and severity (Cunningham et al. 2024; Wang et al. 2025; Wotton et al. 2017). More severe fires in the boreal forest promote species turnover by damaging the seedbed or consuming aerial seedbanks (Johnstone and Chapin 2006; Johnstone and Kasischke 2005; Splawinski et al. 2019), leading to

compositional shifts from well-adapted conifers to faster-growing, and less resilient deciduous broadleaf species (Alfaro-Sánchez et al. 2025; Baltzer et al. 2021; Johnstone et al. 2010). However, detailed understanding of what determines fire severity is challenging. Moisture availability plays an important role due to its effects on fuels, influencing both fuel availability and flammability (Ellis et al. 2022; Jain et al. 2022), but these effects vary spatially and temporally. For example, in humid ecosystems, like the boreal shield east, fuels begin near saturation and respond slowly to increases in VPD, therefore burn severity may not respond as rapidly to sharp increases in VPD, whereas in drier boreal regions, fuels equilibrate rapidly with atmospheric dryness, which should produce faster and stronger short-lag effects on burn severity.

Changes in fuel moisture play a crucial role in determining wildfire burn severity by increasing fuel dryness and the connectivity of flammable biomass, thereby raising the probability of transitions from low to moderate or high-severity fire (Alexander and Cruz 2013; Cruz et al. 2010; Huang et al. 2020; Miller and Urban 2000). Interannual variation in moisture deficits (i.e., long-term moisture availability) is known to contribute to patterns of burn severity from landscape to continental scales (Keyser and Leroy Westerling 2017; Parks et al. 2016; Parks et al. 2014a, b), although this evidence derives primarily from western North American forest systems and direct evidence from eastern boreal forests remains limited. Intra-annual fluctuations in atmospheric moisture conditions (i.e., atmospheric dryness) can also influence wildfire burn severity. Intra-annual increases in fuel dryness and extreme burning conditions can promote high-severity fire (Chavardès et al. 2019; Wu et al. 2018), whereas intra-annual periods of reduced atmospheric dryness (i.e., lower evaporative demand) can promote low-severity fires (Collins et al. 2007; Parks et al. 2018a, b). While other environmental controls on burn severity remain relatively stable over time (e.g., topography, land-cover type), the acceleration of climate change in recent decades has led to high intra-annual variability in atmospheric dryness and interannual variability in moisture availability. Summers are becoming warmer, while the duration of snow cover is decreasing, resulting in earlier snowmelt and warmer spring temperatures (Vincent et al. 2018). Combined, these changes lengthen the fire season and lead to drier conditions, resulting in greater moisture deficits during the fire season that have the potential to alter burn severity.

Studies investigating the time-lagged influence of moisture availability on burn severity typically focus on climate anomalies, that is, the departure from a reference or long-term value or climate normals (Parks et al. 2018a, b, 2014a; Parks and Abatzoglou 2020). Parks et al. (2018a)

used climate moisture deficit (CMD) averaged over 30 years to represent long-term climatic trends driving burn severity, and CMD averaged over the year of fire to represent short-term trends. In this case, short-term climatic trends had a stronger effect on burn severity than longer-term trends, although intra-annual and interannual variability in moisture availability was not explicitly considered. However, because fuel moisture is known to be sensitive to short-term changes in atmospheric conditions (Flannigan et al. 2016), which are seldom considered alongside longer time scales, it is important to examine intra-annual and interannual variability together to develop a comprehensive understanding of what drives wildfire burn severity to inform forecasting of the anticipated future ecological impacts of wildfire.

Here, we examined how short-term atmospheric dryness and long-term climate trends in moisture availability affect wildfire burn severity in the boreal shield ecoregion of northwestern Ontario, Canada. Specifically, we address the following questions: (1) How do forest composition and drought conditions recorded during the active burning period influence burn severity compared to pre-fire moisture availability?; (2) What are the relative effects of short-term atmospheric dryness and long-term climatic trends in moisture availability on wildfire burn severity?; (3) Are the relationships between moisture availability and burn severity consistent for both median and extreme burn severity across fires, reflecting spatial variability in forest composition and climate conditions?; (4) Do lagged moisture effects influence burn severity beyond their average values?; and (5) What are the temporal patterns of lagged moisture effects across their respective windows?

To address these questions, we modelled the effect of short-term (intra-annual) daily lagged atmospheric dryness and long-term (interannual) yearly lagged moisture availability on median and extreme burn severity using distributed lag models (Wood 2017). We also included covariates that capture forest composition (i.e., fuels) and drought conditions over the course of fire duration to assess the relative impact of intra-annual and interannual lagged moisture availability on severity. Here, drought conditions during the fire duration refer specifically to conditions measured from fire ignition date to reported fire end date to capture weather-related fuel moisture conditions that contribute to the majority of burn events over the course of the fire duration and are known to influence fire severity.

As the boreal forest is a seasonally snow-covered region, increasing moisture availability in the winter-spring, long periods of moisture deficit are required to allow large, heavy fuels to dry out enough to increase flammability. Consequently, we expect that short-term

moisture conditions will have the greatest impact on burn severity through their influence on the desiccation rates of fine fuels. Then, as large, heavy fuels make up a significant proportion of available fuels, burn severity is directly related to fuel dryness. We predict, therefore, that (1) short-term moisture conditions (daily, 30 days before fire event) will have a greater influence on burn severity extremes and (2) median burn severity will be influenced by longer-term climatic trends in moisture conditions (yearly, 5 years before fire event).

## Methods

### Study area

Between 2008 and 2018, one-third of the area burned in the boreal forest was in the boreal shield ecozone of Ontario (Coops et al. 2018). The boreal shield ecozone is known for its large, stand-replacing wildfires, with fire return intervals ranging from 50 to 150 years (Hanes et al. 2019; Ter-Mikaelian et al. 2009). We investigated the drivers of burn severity, with specific emphasis on time-lagged moisture availability, in wildfires > 40 ha that burned in the western part of the boreal shield (BSW) ecozone of Ontario (Fig. 2.1), between 1986 and 2024. The western boreal shield is dominated by upland and lowland coniferous and mixed wood forests. The most common tree species are black spruce (*Picea mariana*), jack pine (*Pinus banksiana*), balsam fir (*Abies balsamea*), white spruce (*Picea glauca*), eastern white cedar (*Thuja occidentalis*), and trembling aspen (*Populus tremuloides*) (Ter-Mikaelian et al. 2021). As latitude increases, ground-water and moisture availability also increase as does the frequency of lakes, rivers, bogs, and peatlands.

### Fire and burn severity data

We obtained fire perimeters from the National Burn Area Composite (NBAC) for the years 1986 to 2024 (Skakun et al. 2022). NBAC is a Canada-wide, annually updated fire database that provides maps of areas burned by wildfire by integrating and updating agency fire data with remotely sensed maps dating back to 1972 (Skakun et al. 2022). We focused on fires greater than > 40 ha and excluded all fires that were classified as prescribed burns. Additionally, because climate change is increasing burn rates in boreal forests (Heon et al. 2014), the intensification of droughts and extreme weather conditions is increasing the frequency of short-interval reburns: wildfires occurring in areas that have previously burnt within a short timeframe of years to decades, outside the natural range of historical variability (Whitman et al. 2019). However, burn severity estimates in reburned areas may be biased low, as previous fire severity and post-fire climate alter fuel availability in ways not well captured by remote sensing (Stevens-Rumann et al. 2016; Thompson

et al. 2007). To reduce this bias, we excluded fires that occurred within 15 years of a previous burn; this threshold reflects the earliest available fire perimeter data in the NBAC dataset. In the instances where the intersection of two fires was > 40 ha, we retained the earlier first fire. We chose to use 40 ha as the threshold for intersection between two fires, as the smallest fires in the NBAC dataset are 40 ha.

Using Google Earth Engine (GEE), we estimated burn severity for each fire using the relativized burn ratio (RBR) derived from Landsat surface reflectance imagery, following Parks et al. (2014a, b). Pre- and post-fire composite imagery was drawn from the June–August window in the year before and after each fire. This time period was chosen to capture local phenology, as live deciduous trees and shrubs are expected to have leafed out by June 1 in Ontario's boreal forest (Podur and Martell 2009). RBR was calculated from the normalized burn ratio (NBR) and delta normalized burn ratio (dNBR) using an offset-corrected dNBR to control for variation in unburned areas (see Appendix for equations and full protocol). We used RBR rather than dNBR or RdNBR because it is less sensitive to pre-fire image variability (Parks et al. 2014a, b). Burn severity was summarized at the fire level using the median and 95th percentile RBR, capturing central tendency and upper-bound ecological impact respectively.

### Moisture availability data

Long-term moisture availability was measured using the climate moisture index (Hogg 1997). Climate moisture index (CMI) is a climate-driven index of water balance developed as an indicator of climate dryness to quantify impacts on forests in Canada (Hogg 1997). CMI is calculated as the difference between precipitation and potential evapotranspiration (Hogg 1997), where positive CMI values indicate wet or moist conditions, and negative values signal dry conditions. CMI uses a modified version of the Penman–Monteith equation to calculate potential evapotranspiration (PET) where the landscape for which PET is calculated is assumed to be well-vegetated with adequate soil moisture; therefore, soil and stand level characteristics are not required (Hogg 1997). Across the boreal forest, declines in CMI are positively correlated with annual area burned (Hope et al. 2016), fire size, and fire activity (Dash et al. 2016). CMI is also correlated with boreal forest tree mortality, growth, and overall productivity (D'Orangeville et al. 2016, 2018; Hogg et al. 2005; Wang et al. 2014), all of which are proxies of fuel abundance.

We used the annual average CMI calculated as the sum of monthly values over 12-month periods from August 1 to July 31 of the current year (Hogg et al.

2005, 2013, 2017). Boreal forest tree growth exhibits lagged effects to annual CMI (Itter et al. 2019) and is particularly sensitive to the CMI average over the water year up to 5 years before the year of interest (Hogg et al. 2013; D'Orangeville et al. 2018). The monthly CMI was based on the quantity of monthly precipitation minus monthly potential evapotranspiration, which was computed from a simplified form of the Penman–Monteith equation (Hogg 1997). To explore the potential climate drivers that may contribute to changes in fuel availability and future fire severity, we calculated climate anomalies by subtracting monthly values from 30-year normals of CMI from 1981 to 2010. Climate data were obtained from ERA5-Land climate reanalysis the monthly aggregated via GEE (Muñoz Sabater 2019). CMI anomaly was calculated for each water year for the 5 years before the fire, excluding the year of the fire, and was calculated as the average over the fire perimeter.

Short-term moisture availability was measured by maximum daily vapour pressure deficit. Vapour pressure deficit (VPD) reflects the combined effect of air temperature and relative humidity and is a key factor influencing forest dryness and aridity (Jain et al. 2022). Fire severity is strongly dependent on VPD: at moderate to low VPD, fire severity depends on pre-fire fuel conditions, but when VPD is high, it overrides fuel constraints and leads to extreme severity (Fernández-Guisuraga and Calvo 2025). VPD is coupled with moisture conditions at shorter time scales (Grossiord et al. 2020) and is a good predictor of fuel moisture globally (Rodrigues et al. 2024). VPD is the difference between saturation vapour pressure ( $e_s$ ) and actual vapour pressure ( $e_a$ ) and can be calculated from temperature and dewpoint temperature (Grossiord et al. 2020).

We derived daily maximum air temperature and mean dewpoint temperature time series for each fire perimeter from the ERA5-Land Daily Aggregated-ECMWF Climate Reanalysis (Muñoz Sabater 2019). ERA5-Land Daily Aggregated is a reanalysis product that combines model and observation data to produce daily gridded climate data at 9-km spatial resolution from 1950 onwards. For each fire perimeter, we queried daily maximum air temperature and daily mean dewpoint temperature at 2 m over a 30-day window before the fire event, starting the day prior to fire ignition. Images were spatially aggregated over the fire geometry to obtain one value per day and perimeter. The resulting daily maximum temperature and mean dewpoint temperature were assembled across all fires and days and used to parameterize the inputs for calculating maximum daily VPD from saturation vapor pressure and actual vapor pressure following

established methods (Alduchov and Eskridge 1996; Grossiord et al. 2020; Rothfus 1990; see Appendix for equations). Maximum vapour pressure deficit was summarized as a single value for each day for the day before fire (ignition date –1 day) to 30 days pre-ignition. This allows us to model the lagged effects of VPD on median and extreme burn severity.

### Drought data

To quantify the effect of fire weather on burn severity, we used the drought code (DC) from the Canadian Forest Fire Weather Index System (Van Wagner 1987), which is used operationally by forest managers to predict and evaluate wildfire danger in Canada. DC captures the average moisture content of deep organic layers, evolving slowly to reflect the integrated drying of deep organic and coarse woody fuels under persistent moisture deficits. Unlike CMI and VPD, which characterize atmospheric moisture conditions over interannual and intra-annual timescales respectively, DC provides a fire-behaviour-relevant measure of deep fuel desiccation not captured by meteorological indices alone, allowing us to distinguish the influence of deep fuel drought from pre-fire atmospheric moisture conditions. The drought code has been used previously to successfully model wildfire severity in the boreal forest (San-Miguel et al. 2020; Talucci et al. 2022; Whitman et al. 2018).

We calculated daily DC values from fire season onset to the end of each fire burn period using temperature (°C) and 24 h accumulated precipitation (mm) at noon LST, following Van Wagner (1987) implemented in the *cffdrs* package in R (Wang et al. 2017). Fire season onset was estimated for each fire perimeter at the first date when snow cover dropped below 50% of the burned area for  $\geq 3$  consecutive days, derived from ERA5-Land Daily Aggregated-ECMWF Climate Reanalysis daily snow cover in GEE (Muñoz Sabater 2019). Meteorological inputs were extracted from ERA5-Land hourly data (Copernicus Climate Change Service 2019) at noon LST for each fire perimeter, with UTC offsets applied per fire based on a global time zone shapefile to ensure correct local time alignment. Fire start and end date were drawn from the NBAC, prioritizing hotspot date over agency-reported date where available (Skakun et al. 2022), as hotspot date more accurately reflect the true period of active burning than agency-reported date, which may be subject to reporting delays or administrative timing; fires with missing or identical start and end date were excluded. Default startup values were used for the first day of each DC time series (see Lawson and Armitage 2008 for details). Since the FWI system represents daily fuel moisture and fire potential, while our measure of burn severity is a single value taken at the end of the fire duration, it is possible



that day-to-day variation over the course of the fire influenced the observed burn severity. To better account for this variation in daily fire weather, we calculated the 90th percentile of the drought code over the known duration of the fire. This method allows us to capture only the upper tail of FWI events, which have the most influence on wildfire burn severity (Whitman et al. 2018).

### Forest cover data

We characterized landcover composition within wildfire perimeters using annual forest land cover maps of Canada's entire forested ecosystem for the period 1984–2022 (Hermosilla et al. 2022). These maps provide wall-to-wall land cover characterization at 30-m resolution and were derived from Landsat-based image composites, forest change information, and ancillary topographic and hydrologic data. The classification framework employs regionally tailored models to ensure logical class transitions and reduce year-to-year variability. The dataset represents best-available-pixel composites generated within an August  $1 \pm 30$ -day window each year and has undergone independent accuracy assessment. For methodological details, see Hermosilla et al. (2018, 2022).

Because land-cover estimates might be biased during the fire year due to changes in forest structure following wildfires, we used the preceding year's forest cover map. We assumed that in the absence of fire or another stand-replacing disturbance, forest cover from the previous year is an accurate representation of current year forest cover at the time of fire. If no map was available for the previous year (e.g., 2024), the most recent available image was used. We masked all pixels that were labelled water to ensure that any estimates contained only areas capable of supporting vegetation. We then calculated the percentage of all pixels within a fire that were labelled as coniferous, this excluded mixedwood, broadleaf, wetland treed areas and shrubs.

### Statistical analyses

We assessed the effects of intra-annual and interannual pre-fire moisture availability on burn severity and its effects relative to forest cover and drought conditions during the fire event. We were interested in accounting for the potential delayed effects of moisture availability, but because moisture conditions are likely to be temporally autocorrelated, if models were fit with moisture lags as independent covariates, it would result in overfitting. Instead of fitting multiple models with discrete time lags of moisture conditions as independent covariates, we used a basis function approach (Hefley et al. 2017). Basis functions allow us to represent pre-fire moisture availability as a smooth,

continuous function of time rather than treating each observation at different time lags as separate covariates. Models that incorporate the lags of a covariate using a smoothed function are referred to as distributed lag models (Miller et al. 2025; Wood 2017). This approach captures the underlying temporal structure of moisture dynamics, reduces dimensionality, and mitigates multicollinearity that would arise from including multiple correlated lagged variables (Wood 2017). Essentially, smooth functions summarize the entire moisture history into a set of coefficients associated with smooth basis curves, allowing the model to flexibly approximate lagged effects of moisture over time (Wood 2017).

In addition to accounting for lagged effects over time, we were also interested in capturing any non-linear effects between moisture conditions and burn severity. We expect the effects of both CMI and VPD to be non-linear. For example, at low VPD, fuels dry slowly, but as VPD increases, small increases in VPD can cause fuels to dry rapidly, leading to an exponential increase in flammability, suggesting a threshold response (Burton et al. 2023). Likewise, low CMI may initially increase fire severity by reducing fuel moisture, but prolonged low CMI conditions could increase fuel desiccation and decrease forest growth (D'Orangeville et al. 2018; Mamet et al. 2015; Wang et al. 2014), which may lead to reduced fuel availability, continuity and ensuing burn severity.

Distributed lag models (DLMs) were fit using generalized additive models (GAMs) in the *mgcv* package in R (Wood 2000, 2017, 2025) using a tensor product to create a two-dimensional smooth of a covariate and its lag (Miller et al. 2025; Wood 2017). We fitted separate hierarchical GAMs with distributed lag smooths for pre-fire vapour pressure deficit and climate moisture index anomaly to estimate the effect of moisture availability on either median or extreme burn severity ( $y$ ) as follows:

$$y_i = \beta_0 + f_{cmi}(cmi_i, L_i) + f_{vpd}(vpd_i, L_i) + f_{conf}(coniferous) + f_{dc}(dc_i) + b_{size[i]} + b_{eco[i]} + \varepsilon_i, \quad (1)$$

$$\varepsilon_i \sim Scaled - t_{nu}(0, \sigma^2). \quad (2)$$

where  $y_i$  is either median or extreme burn severity for event  $i$ , and  $f_{cmi}$ ,  $f_{vpd}$  are tensor product (distributed lag) smooths with  $L$  representing the lag index, which is a matrix of 1–5 for CMI and 1–30 for VPD,  $f_{conf}$ ,  $f_{dc}$  are univariate smooth for percent coniferous cover and drought code, respectively and  $b_{size}$ ,  $b_{eco}$  are random intercepts for fire size class (small < 500 ha, medium > 500 ha and < 10,000 ha, large > 10,000 ha, (Stocks et al. 2002)) and ecoregion to account for spatial heterogeneity. We

estimated smoothing parameters by REML and modeled residuals under a Student-*t* family error distribution because residuals were leptokurtotic when a Gaussian family was used.

We fit tensor smooths using penalized thin plate regression splines. In GAMs, the penalty on the smoothed terms is optimized such that more linear shapes are favoured unless the data supports nonlinearity (Wood 2017). We assessed the fit of smooths using estimated degrees of freedom (edf), which shows the complexity of a smooth, where  $\text{edf}=1$  indicates a linearity, and higher values increasing non-linearity. The choice of maximum degrees of freedom is specified by  $k$ , where high values allow for more flexibility. The adequacy of  $k$  for the distributed lag smooth terms was evaluated by extracting model residuals and fitting a model with the single smooth to the residuals while increasing  $k$  and assessing whether increasing  $k$  might explain patterns in residuals. Diagnostic plots were used to assess overall model fit (see Appendix Figures A1, A2).

The relative importance of intra-annual and interannual moisture availability, fuels and drought conditions during the fire event was evaluated by performing hierarchical partitioning of variance. Hierarchical partitioning of variance quantifies each predictor's independent and joint contributions to model fit metrics, in this case, the percent deviance explained (Lai et al. 2022; Peres-Neto et al. 2006). By systematically evaluating all subset combinations of predictors, it provides clear, unbiased estimates of variable importance, enhancing the interpretability of complex non-linear ecological models (Lai et al. 2024). We performed hierarchical partitioning on the fixed-effect predictors (marginal deviance) using the *gam.hp* package in R (Lai et al. 2024). Combined with the overall significance of the individual terms in the model, we used this information to assess whether intra-annual moisture availability (VPD) or interannual moisture availability (CMI) was more important.

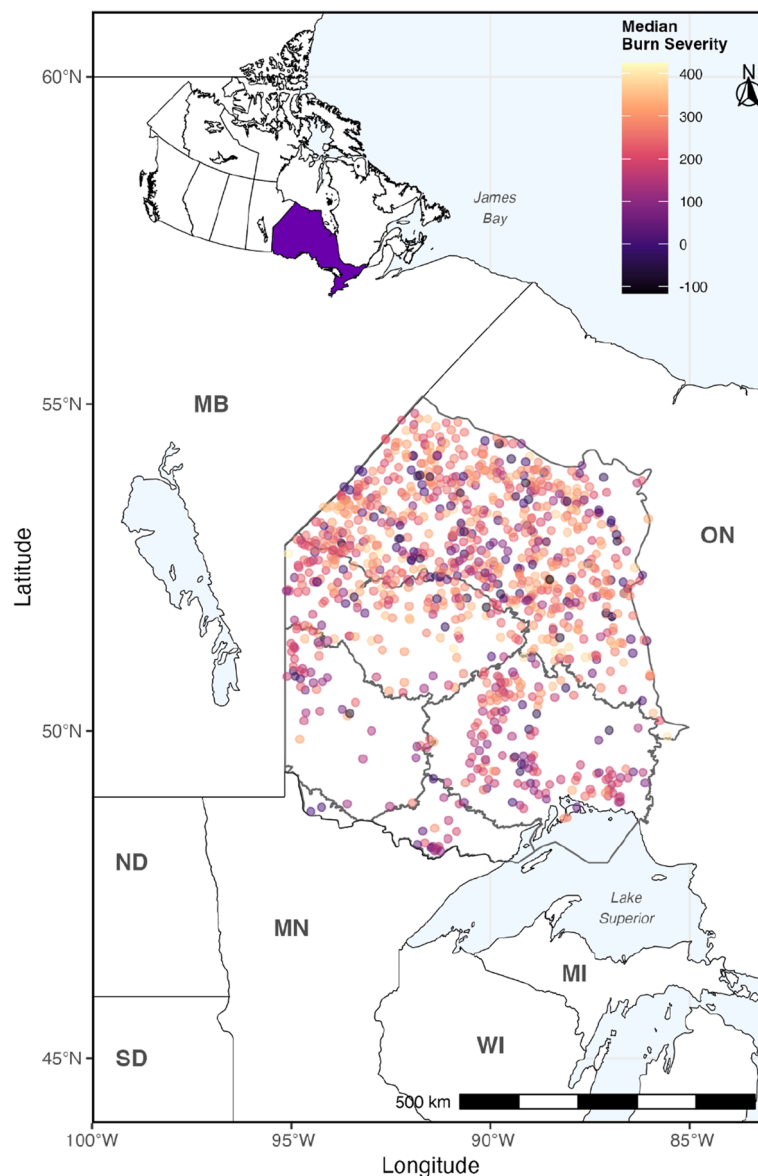
To evaluate whether modelling lagged effects provides explanatory value beyond a simpler summary metric, we compared distributed lag models to models using the mean VPD or CMI over their respective lag windows. Distributed lag models capture variation in influence across time, whereas average-based models assume a cumulative effect. Using likelihood ratio tests (LRT), for each response, we assessed whether the added complexity of modeling temporal heterogeneity significantly improves model fit, thereby determining if the timing of antecedent moisture conditions is more important than their cumulative effect in driving burn severity. To ensure that the likelihood ratio test results were valid, we standardized the smoothing penalties applied to covariates and random effects so that their estimated degrees

of freedom remained comparable across all models. This approach prevents differences in model complexity from biasing the *P*-values and follows best practices for penalized smoothing in GAMs (Wood 2017).

To assess the temporal patterns of lagged moisture effects across their respective windows, we visualized the partial effects of VPD and CMI on each response across its lag dimension. We plotted the partial effects of lagged moisture availability two-dimensional smooths (Fig. 5), and we plotted one-dimensional slices of the tensor smooth across quartiles (Fig. 6). While statistical inference for the overall effect of smoothed terms was conducted using *P*-values, the significant effect of moisture conditions at different lags was assessed by checking for estimates at specific lags where the associated confidence intervals did not cross zero. We highlighted these areas on the two-dimensional smooth plots (Fig. 5). To illustrate uncertainty in one-dimensional slices of the tensor smooths, we drew 200 coefficient samples from the fitted model's sampling distribution (multivariate-*t*), propagated them through the model prediction matrix at each lag to produce an ensemble of partial-effect curves, and summarized the ensemble pointwise by the median and interquartile range while overlaying the single-model fitted curve for reference. For increased clarity of the one-dimensional slice plots, we only included the median of all draws (see Appendix for plots with uncertainty). For CMI, we computed partial effect curves across the 5-year lag window at four quartiles (5%, 25%, 50%, 75%) corresponding to dry-to-wet conditions, while holding other covariates at their median values. For VPD, we computed partial effect curves across a 30-day lag window at the same quartiles (5% to 75%) representing increasing aridity (see Figure A3). This approach allowed us to examine how the influence of antecedent moisture varies over time and across the moisture gradient, highlighting whether certain periods within the lag windows exert disproportionate effects on burn severity. All plots were produced using *ggplot2* (Wickham 2016) and all analyses were conducted in R version 4.5.2 (2025–10–31) (R Core Team 2025).

## Results

We analyzed a total of 1025 fires that burned a total of 2.98 Mha of forested area in the boreal shield west ecozone between 1986 and 2024 (Fig. 1). Most of the burned area occurred in the summer ( $n=973$ ) compared to the spring ( $n=52$ , June 1 cutoff date). Most of the burned area occurred in the Big Trout Lake (BTL) ecoregion ( $n=664$ ), burning a total of 1.81 million ha and accounting for 60.8% of the total area burned. Lake St. Joseph (LSJ) ecoregion accounted for 21.6% of the total burned area, with 149 fires burning a total of 642,957 ha. Lake



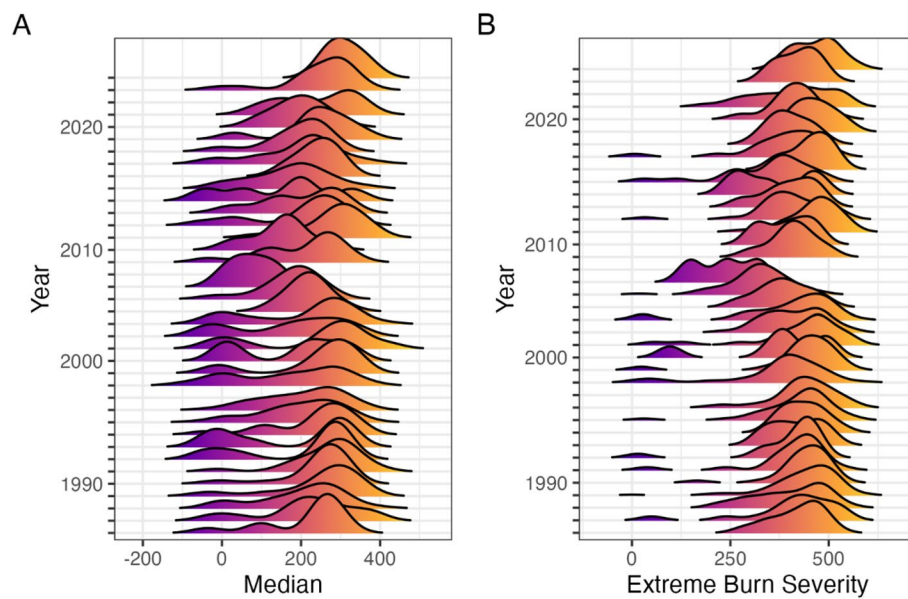
**Fig. 1** Map of the study area in the western Boreal Shield of Ontario, Canada, showing wildfire perimeters  $\geq 40$  ha that burned between 1986 and 2024. Each point represents a fire centroid and is colored by median burn severity. The inset map indicates the location of Ontario within Canada

Nipigon (LN) accounted for 11.6% of area burned, with 146 fires burning a total of 346,092 ha. Lake Wabigoon (LW) and Pigeon River (PR) accounted for 4.68% ( $n=47$ , 149,462 ha) and 1.31% ( $n=17$ , 38,957 ha) of area burned, respectively. Large fires ( $> 10,000$  ha) accounted for 7.32% ( $n=75$ ) of all fires, with medium fires ( $n=484$ , 47.2%) and small fires ( $n=466$ , 45.5%) accounting for most fires.

Fire activity varied throughout our study area with the most active fire years being 1996 ( $n=104$ ), 1989 ( $n=100$ ), and 1995 ( $n=83$ ). Median burn severity was highest in 2024 (RBR=321) and extreme burn severity was highest

in 2019 (RBR=494). Patterns of median burn severity exhibit a broad distribution centered around moderate severity values, but there is substantial interannual variability (Fig. 2a). Higher median severity values are more frequent in recent decades, particularly after 2005, suggesting a shift toward more severe fires over time. Earlier years (1986–1995) show narrower distributions with lower median severity, indicating less variability and generally lower severity (Fig. 2a). Extreme severity values (95th percentile) show pronounced variability across years. High extremes cluster in certain years (e.g.,





**Fig. 2** Annual ridgeline density plots showing the distribution of **A** median burn severity and **B** extreme burn severity for wildfires occurring between 1986 and 2024

mid-1990s, early 2000s, and post-2010), indicating episodic occurrence of very severe fires rather than a steady trend (Fig. 2b). Compared to median severity, extreme severity distributions are more skewed, with occasional years dominated by very high severity events.

### Median burn severity

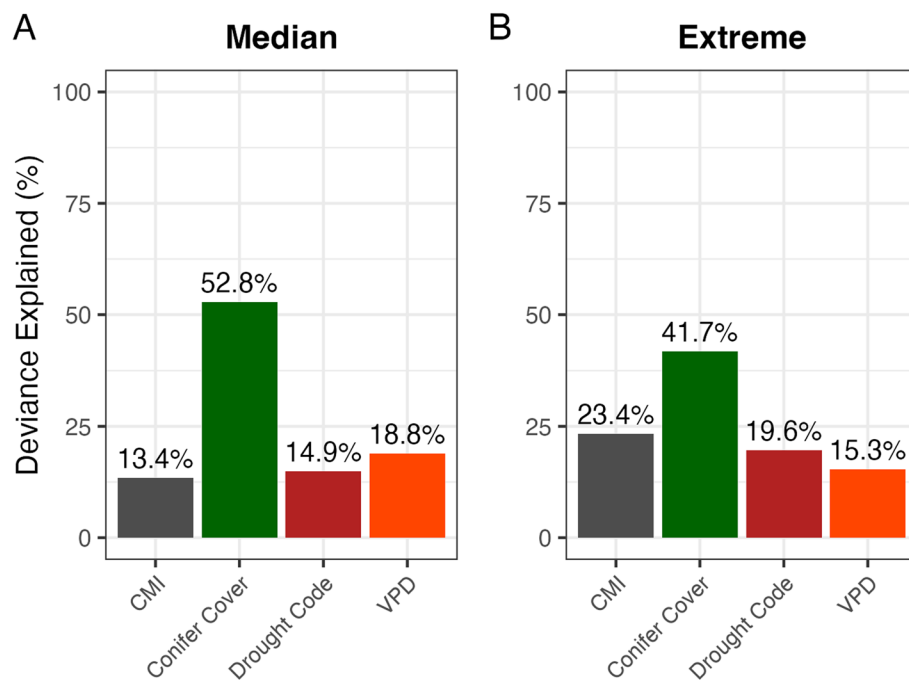
The average relativized burn ratio (RBR) for median burn severity was  $204.02 \pm 20.69$  per fire. The model of median burn severity explained 15.1% of the deviance (adjusted  $R^2 = 0.11\%$ ). To assess the relative influence of short- and long-term moisture availability alongside fuels (i.e., percent coniferous forest cover) and drought conditions during the fire event (90th percentile drought code), we compared their individual contributions to the percent deviance explained by fixed effects using hierarchical partitioning. Percent conifer cover contributed the most to the deviance explained by the fixed effects, accounting for 52.8.2%. Intra-annual moisture availability (VPD) accounted for more deviance explained (18.8.2%, Fig. 3a) compared to both drought conditions over the duration of the fire event (14.9.8%), and interannual moisture availability (CMI), which explained the least amount of deviance (13.4%, Fig. 3a).

We observed a significant effect of percent conifer cover ( $\text{edf} = 2.915$ ,  $\chi^2 = 61.40$ ,  $p < 0.001$ , Table 1), with the partial effect showing a positive, nonlinear contribution to median burn severity across the observed range, with a steepening positive contribution between 50 and 60% conifer cover (Fig. 4a). In comparison, the effect of

drought code was significant, but with a partial effect that was smaller in magnitude relative to percent conifer cover ( $\text{edf} = 5.049$ ,  $\chi^2 = 16.11$ ,  $p = 0.013$ , Table 1), displaying a unimodal pattern whereby the contribution to median burn severity peaked at intermediate DC values before declining at high DC values ( $> 400$ ; Fig. 4b).

Intra-annual and interannual moisture availability had a significant effect on median burn severity. Comparing the distributed lag smooths reveal that their timing and their impact differed. The lagged effect of daily VPD over the 30 days before fire ignition was significant ( $\text{edf} = 2.793$ ,  $\chi^2 = 11.64$ ,  $p = 0.012$ , Table 1). Higher VPD was associated with higher median severity across all lag periods (Fig. 5a). The strongest positive effects were observed at high VPD of ( $> 30$  hPa) at short lags of days 1–6 days before fire ignition (Fig. 5a). At longer lags of 15–20 days before fire, the relationship weakened and transitioned to negative at low-to-moderate VPD ( $\sim 5$ – $15$  hPa), with this negative effect reaching statistical significance (Fig. 5a). A small but significant positive effect was also detected at very low VPD ( $\sim 2$ – $8$  hPa) at approximately 16–19 days lag (Fig. 5a).

In comparison, we also observed a significant effect of CMI that was similar in magnitude to the overall effect of VPD ( $\text{edf} = 9.04$ ,  $\chi^2 = 21.571$ ,  $p = 0.045$ , Table 1). However, the effect of CMI was highly non-linear, relative to a more linear, or additive effect of VPD (Table 1, Fig. 5b). Wetter conditions (high CMI) were associated with lower median burn severity at 1 year lag, whereas dry conditions (low CMI) were associated with increased



**Fig. 3** Hierarchical partitioning of fixed-effects marginal deviance for the median (A) and extreme (B) fire-severity models. Bars show the marginal contribution of each fixed effect to deviance explained: CMI (gray), Conifer cover (green; fuel type), drought code (red), and VPD (orange). Values atop bars are the percent of deviance explained

median burn severity at short lags of approximately 1 year (Fig. 5b). Conversely, dry conditions at 4–5 year lags were associated with decreased median burn severity

**Table 1** Summary of generalized additive model (GAM) results for median and extreme burn severity. Values shown include percent deviance explained for each response, estimated degrees of freedom (edf), test statistics ( $\chi^2$ ), and associated *p*-values for distributed-lag smooths of interannual moisture availability (CMI, lag), intra-annual atmospheric dryness (VPD, lag), and univariate smooths for percent coniferous cover, drought code (DC), fire size class, and ecoregion

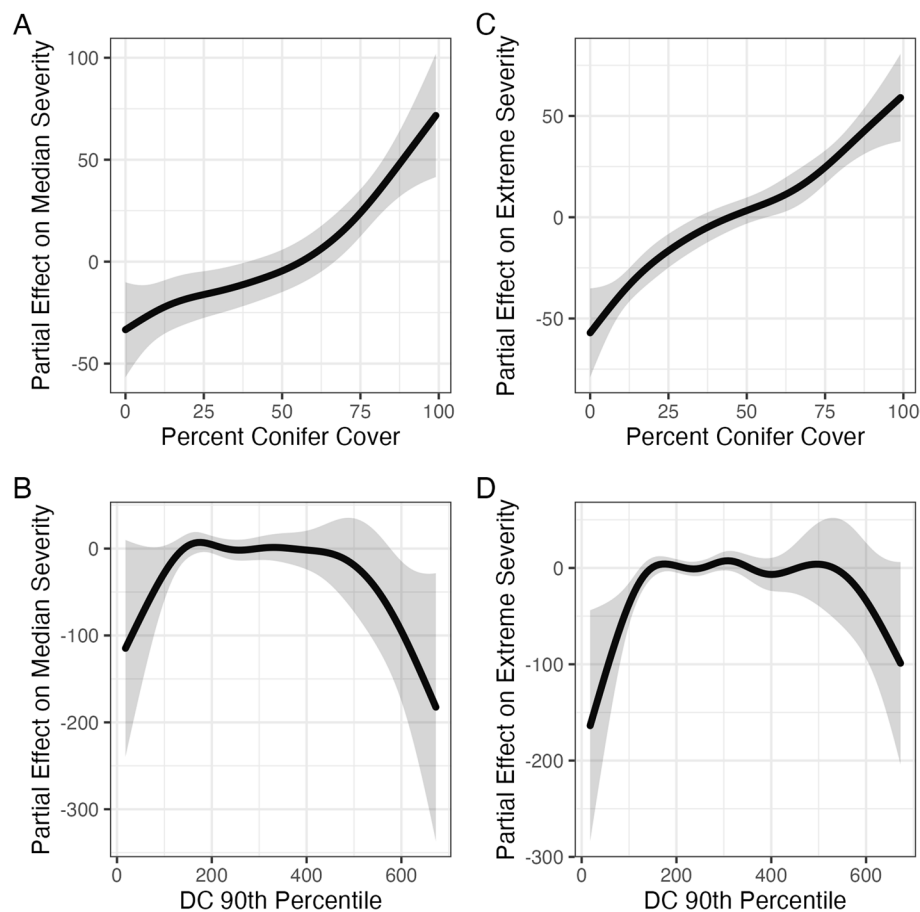
| Response | Deviance explained (%) | Term               | edf    | $\chi^2$ | <i>p</i> -value |
|----------|------------------------|--------------------|--------|----------|-----------------|
| Median   | 15.1                   | te(cmi,Lag)        | 9.0406 | 21.571   | 0.045           |
|          | -                      | te(vpd,Lag)        | 2.7939 | 11.643   | 0.012           |
|          | -                      | s(percent conifer) | 2.915  | 61.400   | <0.001          |
|          | -                      | s(Drought Code)    | 5.049  | 16.115   | 0.013           |
|          | -                      | s(fire size class) | 2.000  | 19.440   | <0.001          |
|          | -                      | s(ecoregion)       | 5.000  | 77.854   | <0.001          |
| Extreme  | 24.5                   | te(cmi,Lag)        | 25.489 | 82.505   | <0.001          |
|          | -                      | te(vpd,Lag)        | 2.300  | 2.320    | 0.401           |
|          | -                      | s(percent conifer) | 3.302  | 145.021  | <0.001          |
|          | -                      | s(Drought Code)    | 6.323  | 26.418   | <0.001          |
|          | -                      | s(fire size class) | 1.853  | 43.357   | <0.001          |
|          | -                      | s(ecoregion)       | 4.448  | 240.599  | <0.001          |

and wet conditions increased burn severity (Fig. 5b). The strongest positive effect was observed at low CMI ( $-28.3$  mm) at 1-year lag, with an estimated  $\sim 35$  unit increase in RBR. The strongest negative effect was also observed at low CMI but at 4-year lag, with an estimated  $\sim 28$  unit decrease in median burn severity (Fig. 5b). No region of the CMI-lag surface reached statistical significance for median burn severity (Fig. 5b).

A likelihood ratio test comparing the model with lagged effects to a model using moving-average summaries of VPD and CMI showed a significant difference between the two model specifications. Holding all other smoothing parameters fixed, including the explicit lagged moisture effects, provided a significantly stronger contribution to model fit (Table 2).

#### Extreme burn severity

The average relativized burn ratio (RBR) for extreme burn severity was  $371.99 \pm 27.04$  per fire. The model of extreme burn severity explained 26.2% of the deviance (adjusted  $R^2 = 0.22$ ). Hierarchical partitioning of fixed effects deviance for the extreme burn severity model suggested that percent conifer cover contributed most to deviance of all the fixed effects in the model (41.7%, Fig. 3b). CMI contributed the second most to fixed effect deviance (23.4%, Fig. 3b); drought conditions through the fire's duration contributed 19.6%, and VPD contributed 15.3% (Fig. 3b).



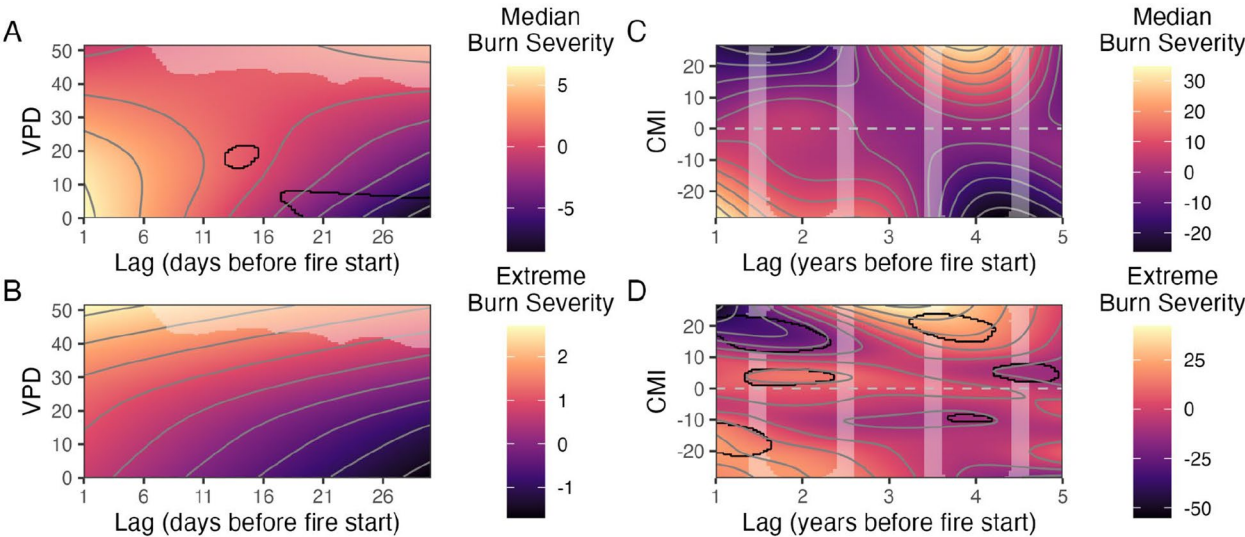
**Fig. 4** Generalized additive model partial-effect plots showing the influence of percent coniferous forest cover (**A, C**) and 90th percentile drought code (**B, D**) on median (**A, B**) and extreme (**C, D**) burn severity. Black curves show fitted smooths, and grey shading indicates 95% confidence intervals

We observed a significant positive effect of percent conifer cover ( $\text{edf}=3.302, \chi^2=145.021, p<0.001$ ). The partial effect shows a general and consistent upward contribution, suggesting an approximately linear relationship between percent conifer cover and its contribution to extreme burn severity (Fig. 4b). In comparison, the DC 90th percentile had a significant effect ( $\text{edf}=6.323, \chi^2=26.418, p<0.001$ ) but the partial effect exhibited a unimodal pattern whereby the contribution to extreme burn severity increased from lower DC values to a peak at intermediate levels before declining at higher DC values (Fig. 4d).

Interannual moisture availability had a significant effect on extreme burn severity ( $\text{edf}=25.489, \chi^2=82.50, p<0.0001$ ), however, the effect of intra-annual moisture availability was non-significant (Table 1). The strongest positive effect occurred at wet conditions (CMI  $\approx +26.6$  mm) at approximately 3–4 year lag, with an estimated  $\sim 42$  unit increase in extreme burn severity (Fig. 5d). The strongest negative effect was also observed

at wet conditions but at 1-year lag, with an estimated  $\sim 55$  unit decrease in extreme severity (Fig. 5d). Dry conditions (CMI  $-21.1$  to  $-11.7$  mm) at 1–2 year lags had a significant positive effect on extreme severity (unit increase of 9.3 to 22.5). A small significant negative effect was also observed under dry conditions (CMI  $-10$  to  $-8.9$  mm) at 3–4 year lags (unit decreases of 10.6 to 12.1). Near-neutral moisture conditions (CMI  $+1.7$  to  $+5.6$  mm) at 1 to 2 year lags had a significant positive effect (unit increases of 8.9 to 13.5). Similarly, a negative effect occurred at CMI  $+2.8$  to  $+7.2$  mm at 4–5 year lags (unit decreases of 8.2 to 13.4). Wet conditions (CMI  $+12.2$  to  $+23.3$  mm) at 3–4 years had significant positive effects (unit increase of 16.8 to 32.8). The effect of daily VPD on extreme burn severity was nonsignificant ( $\text{edf}=2.300, \chi^2=2.32, p=0.4$ ), though the surface showed a weak positive tendency at high VPD and mid-range lags and a weak negative tendency at low VPD across longer lags (Fig. 5c).

Model comparison using a likelihood ratio test showed that replacing the lagged-effects smooths



**Fig. 5** Distributed-lag effect surfaces showing the modeled influence of intra-annual VPD (**A, B**) and interannual CMI anomalies (**C, D**) on median and extreme burn severity. The x-axes represent lag time prior to fire (days for VPD; years for CMI), and colors indicate the estimated partial effect magnitude. Black contours outline regions where the 95% confidence intervals of the smooth term do not overlap zero, marking statistically credible lagged effects. Gray regions identify areas outside the model's data-supported prediction envelope, locations in lag–covariate space with insufficient observations or where GAM predictions would rely on extrapolation and thus are not interpreted

**Table 2** Likelihood-ratio tests comparing models using lagged versus averaged moisture predictors for median and extreme burn severity. Higher LRT values indicate better performance of the more complex (lagged) model relative to its simpler counterpart. Lagged models incorporate distributed-lag smooths for moisture availability, whereas averaged models use moving-average summaries over the same lag windows

| Models compared                     | Response | LRT <sup>1</sup> | $\Delta df^2$ | <i>p</i> |
|-------------------------------------|----------|------------------|---------------|----------|
| Lagged vs. Average                  | Median   | 46.45            | 18.71         | < 0.001  |
| Lagged vs. Average                  | Extreme  | 104.55           | 41.40         | < 0.001  |
| Lagged vs Lagged CMI + VPD Averaged | Extreme  | 103.90           | 33.73         | < 0.001  |

<sup>1</sup> LRT = chi-square test statistic for the likelihood-ratio test:  $-2(\text{LogL}_{\text{Averaged}} - \text{logL}_{\text{Lagged}})$   
<sup>2</sup>  $\Delta df$  = degrees of freedom for the  $\chi^2$  test statistic defined as the difference in the number of (co)variances fitted for the 2 models

with a 5-year moving average significantly reduced model fit (Table 2). When both lagged smooths were replaced with moving averages, 30-day average VPD became significant ( $z = 1.9858$ ,  $p = 0.012$ ) while the 5-year CMI average was non-significant, suggesting that the moving averaged VPD was capturing variance better explained by the lagged CMI in the more complex model. We therefore refit the model retaining the lagged CMI smooth while replacing the lagged VPD with the 30-day moving average in this model VPD was non-significant, consistent with the lagged CMI accounting for the shared variance between the two

predictors. Despite VPD being non-significant, this model provided the best overall fit (Table 2).

Discussion

Our analysis of burn severity for 1025 wildfires across more than three decades in the boreal shield west reveals consistent patterns in how antecedent moisture conditions, fuels, and fire-weather during the active burning period shape variation in both median and extreme burn severity. Across all models, forest composition and pre-fire moisture availability emerged as dominant influences, with coniferous cover exerting strong, positive effects and moisture indices contributing substantially to explain deviance at both intra-annual and interannual time scales (Fig. 3). Contrary to expectations, long-term, interannual moisture conditions (CMI) exerted stronger effects across both median and extreme severity relative to short-term, intra-annual conditions (VPD). However, their combined contributions exceed the influence of moisture conditions during the active burning period. Distributed-lag analyses showed that moisture effects were not uniform across time but instead varied with lag structure, producing distinct windows in which moisture anomalies corresponded to either elevated or reduced burn severity outcomes. These lagged responses contrasted with the moving-average summaries, which provided more spatially consistent linear effects for short-term moisture (VPD), and showed weak support for long-term moisture (CMI). Model comparisons



further indicated that, for extreme severity, the cumulative effects of intra-annual moisture are more impactful. Together, these results underscore the complexity of climate–fuel interactions governing burn-severity patterns and highlight the value of considering both long- and short-term moisture metrics, as well as their temporal structure, when evaluating fire-severity drivers across heterogeneous boreal landscapes. Our findings are directionally consistent with studies from temperate and boreal forest regions, primarily in western North America, which have shown that wildfire burn severity is sensitive to pre-fire moisture availability at both short- and long-term time scales (L. Collins et al. 2022; Parks and Abatzoglou 2020; Wasserman and Mueller 2023; Whitman et al. 2018). Direct evidence from eastern boreal systems remains limited, and Goldman et al. (2025) represents one of the few studies to examine these relationships in a comparable Canadian context. To our knowledge, previous studies have not directly compared short intra-annual variability and long interannual variability in moisture availability within a single analytical framework.

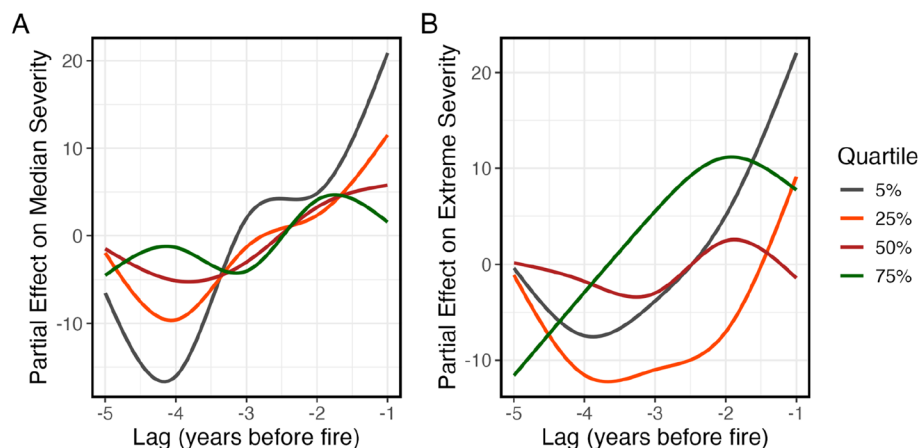
The strong contribution of pre-fire moisture availability to both median and extreme burn severity indicates that fires in the boreal shield west are strongly preconditioned by moisture history operating across multiple timescales rather than simply governed by fuel composition at ignition. In Ontario's boreal shield west, prolonged drought-like conditions prior to fire contribute to more extreme fire events (Beverly and Martell 2005), and long-term warming trends are positively correlated with summer fire severity (Shabbar et al. 2011), reflecting the accumulation of seasonal and interannual moisture deficits that elevate fuel desiccation and ignition potential over time. Although the drought code tracks moisture from fire season onset and therefore reflects both pre-fire and contemporaneous conditions, summarizing DC as the 90th percentile over the known fire duration captures a distinct temporal window from VPD and CMI. The modest contribution of DC relative to VPD and CMI (Fig. 3) suggests that moisture anomalies operating at intra-annual and interannual timescales exert a stronger influence on burn severity than cumulative seasonal during conditions spanning the fire duration.

Although previous studies have shown relationships between high-severity fire and fire weather indices in this region (Goldman et al. 2025), the relative importance of fire weather on burn severity is mixed in the literature. While some studies find fire weather highly influential (Keyser and Leroy Westerling 2017; Wang et al. 2022, 2025), others suggest a marginal impact relative to climate and pre-fire conditions (Birch et al. 2015; Parks et al. 2018a, b). Many of these studies span broad geographic

extents across multiple provinces or ecozones, where spatial aggregation can smooth local heterogeneity. For example, Wang et al. (2025) identified BUI as the most influential predictor across the full BSW extent from western Ontario to northern Alberta, whereas our analysis within a single provincial boundary is better positioned to detect finer-scale drivers that may be obscured at broader extents. While fire weather can influence fire behaviour, its role in driving high-severity fire may be secondary to cumulative moisture deficits and long-term climatic trends. As warming continues and antecedent drying intensifies in the boreal forest (Wang et al. 2014), the relative importance of pre-fire conditions is likely to increase, shaping future fire severity patterns across the boreal shield.

The strongest positive effects of VPD on median burn severity were observed at short lags (1–6 days), consistent with the well-established role of acute atmospheric moisture demand on fire through the rapid drying of fine fuels in the days immediately preceding ignition (Seager et al. 2015). Fuels of different species, sizes, and structures dry at different rates, ranging from hours to days for fine fuels and extending to weeks for larger fuels and deeper organic matter (De Magalhães and Schwilk 2021; Lawson and Armitage 2008; Skinner et al. 2006). Coarse woody fuels such as snags, logs, and other dead debris represent key contributors to high-severity fire, as their large fuel loads and structural properties markedly enhance fire severity (Lydersen et al. 2019). Because VPD-driven drying often exerts its strongest influence at weekly scales (Grossiord et al. 2020), the significant negative effect of low-to-moderate VPD (~5–15 hPa) at 15–20 day lags may reflect the slower drying dynamics of larger fuels and organic matter, whereby periods of reduced atmospheric demand in the weeks prior to fire maintain higher moisture content in these fuel classes, moderating burn severity when fire occurs.

The linear effect of 30-day mean VPD on extreme burn severity was stronger than its potential non-linear lagged effects within the same pre-fire window. Because fuel desiccation is a cumulative process, where consecutive days of elevated atmospheric dryness progressively reduce moisture across fine and coarse fuels, multi-day summaries may provide a more ecologically meaningful measure of drying than single-day conditions. Moving averages capture this cumulative influence by integrating sustained VPD into a single metric. This interpretation is supported by regional evidence showing that long dry spells exceeding 15 days are major drivers of extreme fire events in the boreal shield west (Beverly and Martell 2005). It also aligns with broader studies demonstrating strong relationships between cumulative VPD and increased fire severity (Seager et al. 2015; Sedano and



**Fig. 6** Partial-effect curves extracted from one-dimensional slices of the distributed-lag CMI smooth, showing how moisture anomalies over the preceding five years influence median (left) and extreme (right) burn severity. Each curve represents a different position along the moisture gradient from low-to-high (5%, 25%, 50%, and 75% quartiles), with other covariates held at median values. The curves shown are the median of 200 posterior draws propagated through the model prediction matrix; the full uncertainty is presented in Appendix Figure A6. These slices reveal distinct lag-dependent and moisture-dependent patterns, highlighting specific periods where antecedent climate exerts stronger positive or negative effects on burn severity

Randerson 2014; Williams et al. 2014). This is particularly relevant in mesic forests like those found in boreal shield west, where a VPD threshold of approximately 55 hPa is necessary for higher-severity fires (Cawson et al. 2024); the highest observed VPD in our study was  $\sim 52$  hPa, suggesting that acute single-day VPD rarely reaches levels sufficient to drive extreme severity in this region, and that cumulative atmospheric drying capture by the 30-day average is therefore the more ecologically meaningful metric. Overall, median severity was driven by lagged moisture history, while extreme severity reflected a distinct response to sustained atmospheric drying, highlighting that typical and extreme fire outcomes arise from different underlying processes.

Interannual moisture availability exerted a strongly non-linear and lag-dependent influence on burn severity in the boreal shield west. For both median and extreme burn severity, CMI effects showed a temporal reversal reflecting fundamentally different influences of prior wet and dry conditions across short and long lags (Fig. 6a, b). At 1–2 year lags, drier-than-normal conditions (low CMI) were associated with increased severity and wetter-than-normal conditions (high CMI) with decreased severity (Fig. 6a, b), while at longer lags (3–5 years) this relationship reversed (Fig. 6a, b). Wetter-than-normal conditions several years before fire can enhance vegetation growth and fuel availability, which may subsequently elevate burn severity once those fuels cure, supported by evidence that boreal tree growth responds non-linearly to CMI across the full compositional gradient of eastern Canadian boreal forests (D’Orangeville et al. 2018). The

net accumulation of organic layer fuel loads in Canadian boreal forests reflects the balance between litterfall inputs and decomposition, both of which are sensitive to interannual moisture availability, suggesting that CMI may influence not only vegetation growth but the available surface fuel load at the time of burning (Hanes et al. 2022).

Although boreal fire regimes are broadly characterized as moisture-limited rather than fuel-limited (Meyn et al. 2007; Whitman et al. 2018), these paradigms are not mutually exclusive. Meyn et al. (2007) acknowledged that antecedent moisture can modulate fire outcomes on interannual timescales as a secondary mechanism. The lagged positive relationships we observe are consistent with this secondary mechanism, whereby wet years preceding fire may amplify burn severity by promoting understory and shrub growth where water availability is the primary constraint on understory productivity (Chen et al. 2025), facilitating the accumulation of ladder fuels that connect surface fuels to the canopy (Flannigan et al. 2016), lowering functional canopy base height, reducing the threshold for crown fire initiation. Together, these mechanisms suggest that the legacy of antecedent moisture shapes the specific stand structural attributes that are proximate drivers of crown fire initiation and spread, and therefore of remotely sensed burn severity.

In boreal forests, increases in CMI correspond to higher productivity and tree growth, across the full compositional gradient of eastern Canadian boreal forests (D’Orangeville et al. 2018). While years with above-average moisture availability are generally considered

beneficial for reducing wildfire risk, our results reveal a potential lag-dependent positive feedback between antecedent moisture surplus and burn severity. When conditions are drier-than-normal, signalling potential drought-like conditions, tree mortality or water stress may decrease the amount of live fuel and increase fuel availability, contributing to higher-severity fires. Importantly, and independent of the growth-to-fuel pathway, the eastern boreal forest exhibits long ecological memory to moisture deficits up to 6 years, with moisture deficits a longer 2–3 lags exhibiting greater importance on growth than deficits in immediately preceding year (Itter et al. 2019). This temporal structure of lagged moisture responses in boreal tree growth is consistent with multi-year lag windows we observe in our distributed lag models, suggesting that the influence of antecedent moisture on burn severity operates through cumulative, time-delayed processes rather than immediate responses. Together, these patterns demonstrate that interannual moisture variability governs fuel availability over multi-year timescales, meaning that the legacy of past moisture conditions, not just conditions in the fire year, plays a critical role in shaping burn severity.

The strength and the consistency of the relationship between percent conifer cover and median and extreme burn severity further underscore the importance of fuel availability as a driver of burn severity. Within the moisture-limited framework established above, conifer cover still constrains the upper bound of burn severity at the fire level, consistent with evidence that fuel structure modulates the climatic threshold at which vegetation becomes highly flammable (Pausas and Paula 2012, Fernández-Guisuraga and Calvo 2025). However, this relationship may change as broadleaf species are becoming more flammable under climate change (Parisien et al. 2023). We did not explore the potential for interacting effects between percent conifer cover and interannual or intra-annual moisture availability. Drier-than-normal conditions in preceding years may increase the flammability of broadleaf species and alter the nature of the relationship between conifer cover and burn severity. Given that CMI-related moisture deficits are heavily linked to aspen growth declines across the boreal forest (Anderegg et al. 2013; Hogg et al. 2005), it is plausible that the effect of CMI was more pronounced across the range of conifer cover examined in our study, and may explain why VPD was less influential. The relationship between interannual moisture availability and severity in fires that burned in predominantly broadleaf or mixedwood areas warrants future analysis.

## Limitations

These results should be interpreted within the context of the specific ecoregions, climate regimes, and datasets used. The boreal shield west occupies subhumid to moist high- and mid-boreal ecoclimatic zones, characterized by cool summers, long winters, short growing seasons and low annual precipitation (Ecoregions Working Group 1989). This cold, moisture-limited climate amplifies the role of antecedent CMI in explaining burn severity, as fuel availability is strongly shaped by interannual moisture variability. In wetter regions or more productive regions, fuel loads may remain high regardless of moisture anomalies, shifting burn severity control towards in-season weather. Ecosystems with faster decomposition rates or different vegetation structures may similarly not exhibit the same sensitivity to multi-year moisture history, limiting transferability beyond moisture-limited boreal systems.

Our conclusions also depend on the variables included in our models. While CMI captures broad climate-driven fuel variability, finer-scale attributes, such as canopy density, coarse woody debris, understory composition, and stand-level structural were excluded because they are not comprehensively mapped across our study domain and operate at finer resolutions than our regional analysis. Including such metrics could increase the explanatory power of fuel-driven processes or alter the relative importance of lagged moisture. Additionally, because CMI functions at landscape scales while many fuel-accumulation processes act at stand or microsite scales, uncertainty remains in how well climate anomalies reflect actual ground-level fuel loads. Our findings should therefore be viewed as region-specific to Ontario's boreal shield west rather than universally applicable.

A further limitation concerns the DC. Its modest contribution and unimodal partial effect may reflect our model's omission of the overwintering effect, a process of growing importance in northern boreal regions where deep organic layers retain moisture deficits from the preceding fall (Hanes et al. 2020). Without overwintering, DC resets annually, preventing capture of carryover effects that could elevate burn severity. This omission may also partly explain the strong 1-year lagged CMI effect, as CMI could be indirectly proxying persistent moisture conditions otherwise captured by an overwintered DC. Future work incorporating overwintering DC or continuous multi-year soil-moisture indices would help clarify whether the observed unimodal response reflects true ecological behaviour or a methodological constraint.

## Conclusion

We found that pre-fire moisture conditions driven by interannual variability in moisture availability contribute significantly to wildfire burn severity. Consistent with previous work, fuel availability also exerts strong control over burn severity in Ontario's boreal shield west. Importantly, we identified lagged effects which antecedent moisture conditions can either amplify or dampen future burn severity. Wetter-than-normal conditions 4–5 years before fire increase severity, while drier-than-normal conditions appear to reduce burn severity, likely due to fuel limitations. Because fuel limitations strongly influence wildfire occurrence and severity (Coops et al. 2018), longer-term shifts in moisture availability under climate warming may increasingly constrain future burn severity.

At longer time lags, however, additional factors may also shape burn severity. Our results provide a basis for understanding present and future burn-severity patterns under changing water availability, weather, and climate. Burn severity has important implications for forest resistance, resilience, recovery, and successional trajectories. High-severity fire can slow recovery, reduce resilience, and shift successional pathways, with consequences for biodiversity and ecosystem services. Therefore, anticipating and predicting burn severity will be essential for effective forest management and conservation planning. Future work is needed to clarify how climate-driven changes in fuel abundance and fuel moisture interact to influence wildfire severity.

## Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s42408-026-00544-x>.

Additional file 1: Supplementary material.

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## Authors' contributions

JAG: Conceptualization, data curation, formal analysis, investigation, methodology, project administration, validation, visualization, writing – original draft; PMAJ: Conceptualization, methodology, funding acquisition, supervision, writing – review and editing; MJF: Methodology, funding acquisition, supervision, writing – review and editing.

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## Data availability

All data used in this manuscript are publicly available. The wildfire disturbance area polygon data is available through the Government of Canada's National Burned Area Composite data product <https://cwfis.cfs.nrcan.gc.ca/datamart/metadata/nbac>. Forest structure data are available via the Government

of Canada Open Data Portal: <https://open.canada.ca/data/en/dataset/ec9e2659-1c29-4ddb-87a2-6aced147a990>. Climate data used to calculate climate moisture index from ERA5-Land Monthly Aggregated – ECMWF Climate Reanalysis are available in Google Earth Engine's data catalog: [https://developers.google.com/earth-engine/datasets/catalog/ECMWF\\_ERA5\\_LAND\\_MONTHLY\\_AGGR](https://developers.google.com/earth-engine/datasets/catalog/ECMWF_ERA5_LAND_MONTHLY_AGGR). Weather data used to calculate the drought code are from ERA5-Land Daily Aggregated – ECMWF Climate Reanalysis are available in Google Earth Engine's data catalog: [https://developers.google.com/earth-engine/datasets/catalog/ECMWF\\_ERA5\\_LAND\\_DAILY\\_AGGR](https://developers.google.com/earth-engine/datasets/catalog/ECMWF_ERA5_LAND_DAILY_AGGR). The data supporting this article and code are available on GitHub: <https://github.com/jackagoldman/lagged-moisture-severity>.

## Declarations

### Competing interests

The authors declare no competing interests.

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